

Vehicle Re-identification method based on Swin-Transformer network

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ABSTRACT

Vehicle Re-identification is to find out the exact vehicle captured by other cameras given a vehicle image. In natural traffic surveillance systems, vehicle re-identification can play a role in target vehicles localization, supervision, and criminal investigation. While current convolutional neural network-based approaches have been very successful, most studies consider learning representations from individual images, ignoring potential interactions between them. The Swin Transformer method currently shows excellent advantages in image classification. It can handle features at different scales like convolutional neural networks and make elements more closely related to each other, enhancing the learning of robust features and improving feature robustness and discrimination compared to convolutional neural nets. Therefore, this paper adopts the Swin Transformer method for vehicle re-identification and conducts experimental validation on two large vehicle datasets, VeRi-776, and VehicleID. The experimental results show that the recognition effect of this method is significantly better than existing vehicle re-identification methods, such as VAM, FDA-Net, TransReID, etc.

1. Introduction

With the increasing number of vehicles in modern society, various traffic problems have arisen. In this situation, vehicle re-identification is gradually being widely studied. The vehicle re-identification problem is the retrieval problem of determining whether the vehicle images captured in a non-overlapping area belong to the exact vehicle in a specific range of traffic monitoring scenes. Its purpose is mainly to match the vehicle images with the captured vehicle images by the camera. However, the need to perform vehicle searches from gallery sets has increased due to surveillance cameras and traffic management on smart city roads. Vehicle re-identification is similar to several other applications, such as person re-identification, behavior analysis, cross-camera tracking, vehicle classification, object retrieval, object recognition, etc. Currently, surveillance cameras are installed in traffic-developed areas at home and abroad. Studying how to use them for traffic supervision and a criminal investigation is a fascinating problem.

Before the emergence of deep learning techniques, early research on vehicle re-recognition focused on manually designing better visual features and learning better similarity metrics. In recent years, deep learning techniques have been widely applied to vehicle re-identification tasks with the development of deep learning. Unlike traditional methods, deep learning methods can automatically extract better vehicle image features and learn better similarity metrics

simultaneously. Object Re-Identification aims to correlate specific objects across scenes and camera views. Removing robust and distinguishable features is an integral part of Re-ID and has long been dominated by convolutional neural network (CNN) based approaches [1–3].

Despite the great success of methods based on convolutional neural networks (CNN) [4], they process one local neighborhood at a time and suffer from a loss of detailed information caused by convolution and downsampling operators such as pooling and stepwise convolution. CNN works on edges, shapes, and original vehicle features but does not consider the relationships between these features; therefore, when vehicle images are rotated or captured in a different rotation are caught, the model's performance is often unsatisfactory. On the other hand, the advent of the attention mechanism essentially solves the problem of detail loss in convolutional neural networks.

In natural language, Transformer [5] has become its preferred model. Transformer outperforms RNN and CNN on machine translation tasks. Transformer is a model with an attention mechanism. In biological terms, the attention mechanism can be viewed as a unique brain signal-processing mechanism of the human visual system. When a picture appears in front of the eyes, the human vision quickly skims the whole picture and filters out the vital target regions. In the field of computer vision inspired by a Transformer, Google proposed the landmark model ViT (Vision Transformer) [6]. Its primary design is to apply

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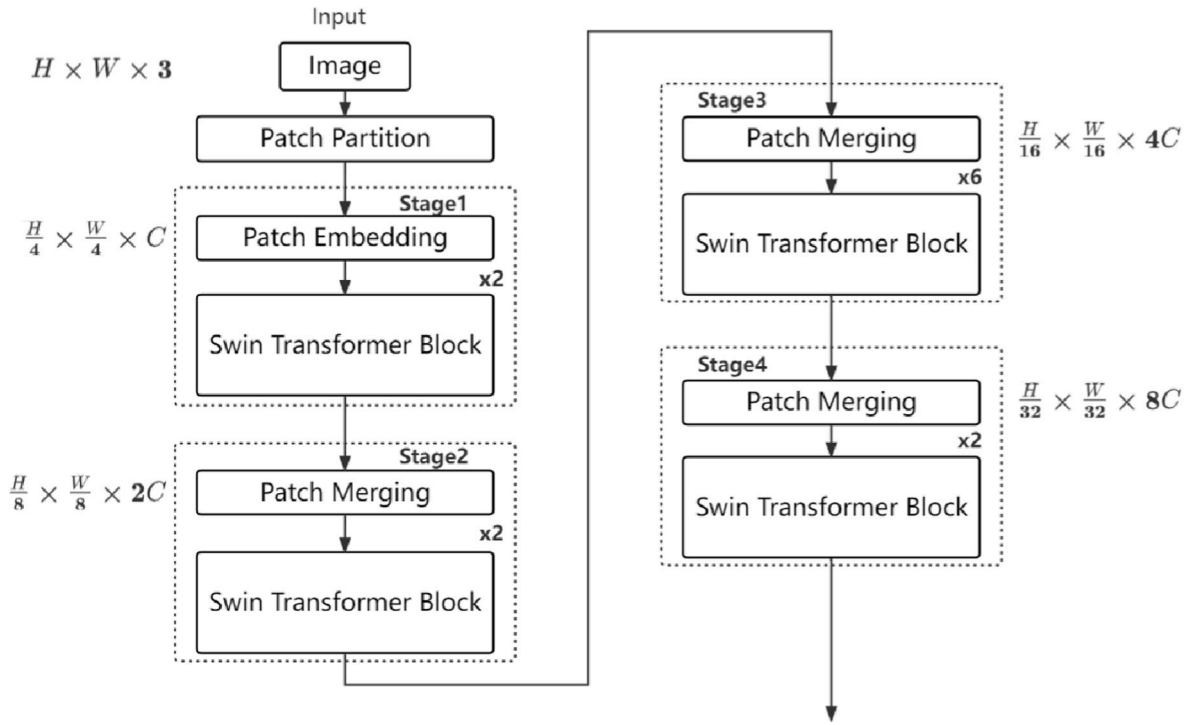


Fig. 1. Swin transformer model.

a pure Transformer architecture directly to a series of image blocks for classification tasks, which can achieve excellent results. It also outperforms state-of-the-art convolutional networks on many image-classification tasks. In 2020, the Transformer model was first applied to an image classification task and obtained better results than the convolutional neural network model. Since then, many researchers have tried to use the powerful modeling ability of the Transformer model in the field of computer vision.

Currently, the Transformer-based framework is more often used for image classification. Transformer models have become a new paradigm in the natural language processing field. Nowadays, more and more researchers are trying to apply the powerful modeling capabilities of Transformer models to computer vision. In Vehicle re-identification, CNN involves to edge, shape, and original vehicle features but does not consider the relationship between these features. And the emergence of ViT solves this problem very well. It performs well in dealing with different poses, orientations, and obscured objects. ViT (Vision Transformer) and DeiT (Data efficient image Transformers) [7] proved that in feature extraction for image recognition, a Transformer can be based on a convolutional neural network-based approach is as practical.

TransReID [8] is an approach to vehicle re-identification through the ViT model, proposing to rearrange patch embeddings through shift and patch blending operations to produce robust features with improved discrimination and more diverse coverage. The patch size inside the ViT model is fixed and is a uniform scale. For the natural language processing domain, the scale is uniform, but in the computer vision domain, the patch size of the graph is not fixed and can be large or small. When performing some classification tasks, the patches may be more significant. But for downstream tasks in computer vision, e.g., target detection, pixel-level segmentation or generation, etc., the patch size usually needs to be adjusted. This may lead to computational problems for ViT, and if the patches are constant, it is not a good solution for these downstream tasks. The emergence of the Swin Transformer is a solution to the problem that ViT is not friendly for downstream tasks.

Swin Transformer [9] solves this problem and achieves better results in classification, detection, and segmentation tasks. Most of the hyperparameters commonly found in convolutional neural networks are also

manually adjustable in Swin Transformer, such as the number of network blocks that can be adjusted, the number of layers in each block, the size of the input image, and so on.

Before Swin Transformer, ViT and iGPT used small-size images as input, and this strategy of directly resizing images would undoubtedly lose much information. Unlike them, the input of Swin Transformer is the original image size, e.g., 224*224 for ImageNet. Swin Transformer uses the most common hierarchical network structure in convolutional neural networks. This feature is also satisfied in Swin Transformer.

Swin Transformer is a modification of ViT using sliding windows. It divides the fixed-size sampling blocks in ViT into windows of different sizes according to the hierarchy, completes the calculation of the self-attention mechanism of the pixel points of the windows, and realizes the information exchange of the pixel points between different windows through the sliding window mechanism. Compared with the application of the TransReID method in vehicle re-identification, Swin Transformer expands the network's "field of perception" and improves the efficiency of information utilization. Therefore, this paper uses Swin Transformer's method for vehicle re-identification. The structural arrangement of this paper mainly consists of an introduction, Swin Transformer model description, experimental validation, and a conclusion. In summary, it mainly contains the following two aspects.

- (1) In this paper, the Swin Transformer model is used as the backbone network to extract the global feature map of vehicle images, and the similarity measure with the exact location of vehicle features is implemented to improve the accuracy of the vehicle re-identification algorithm.
- (2) Experimental results on two large vehicle datasets (VeRi-776 and VehicleID) show that the performance of the employed method outperforms that of most state-of-the-art vehicle re-identification methods.

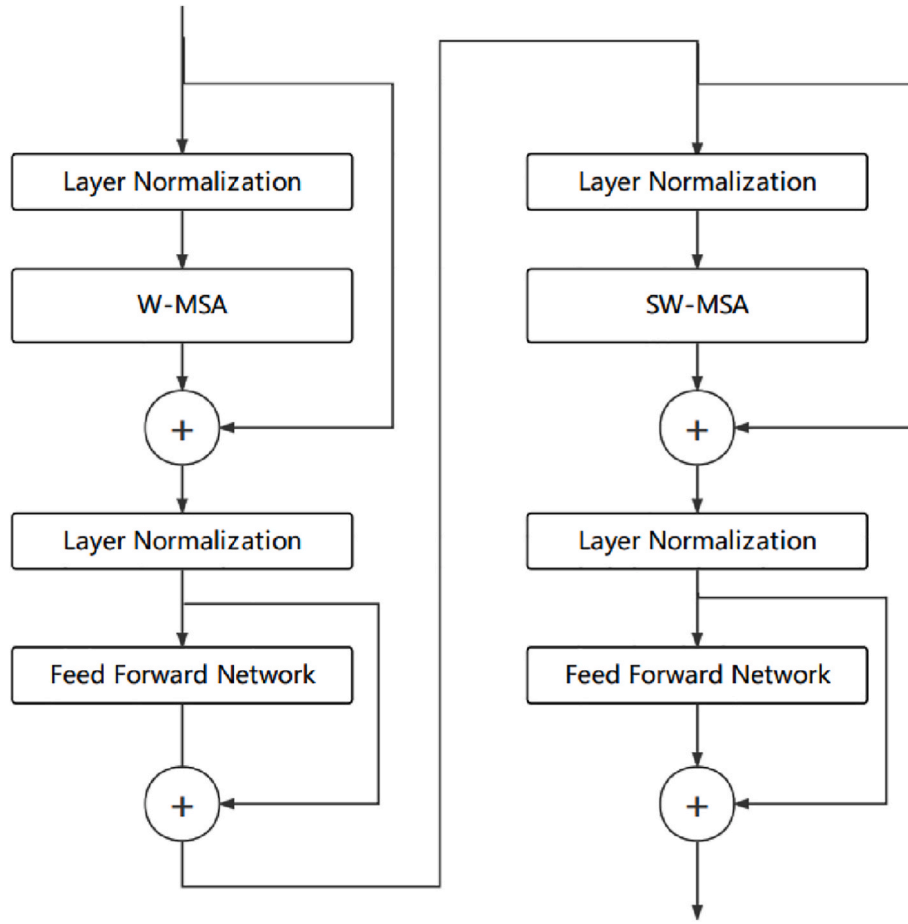


Fig. 2. Swin transformer block modules.

2. Swin Transformer model

2.1. Swin Transformer Model architecture

Swin Transformer is based on the ViT using sliding windows. The architecture of the Swin Transformer is identical to that of the ViT, as shown in Fig. 1. These two modules are roughly the same as those in ViT. Still, the Multi-Head Attention mechanism (MSA) is replaced by W-MSA (Windows Multi-Head Self Attention) and SW-MSA (Shifted Windows Multi-Head Self Attention), which adopt the delimited window and sliding window. The multi-head attention is achieved by using a delimited window and a sliding window to improve computational efficiency and ultimately classification accuracy.

Input an image, $x \in \mathbb{R}^{H \times W \times C}$, where H , W and C represent the image's height, width, and the number of channels of the image, respectively. This is consistent with the division of patch sizes in the ViT framework. Use the sliding window mechanism to slide the image x to divide the partially pixel-overlapping patch blocks. The input image x with resolution $H \times W$ is divided into N fixed-size patch blocks with the following Formula (1).

$$N = N_H * N_W = \left\lceil \frac{H + S - P}{S} \right\rceil * \left\lceil \frac{W + S - P}{S} \right\rceil \quad (1)$$

Where N_H and N_W represent the number of Patch blocks on the image x height and width, S is the step length of the slide, and P is the edge length of the patch.

Each patch is also called a token, and the pixels of each token are RGB tricolor channels. In the implementation, the size of the divided patches is 4×4 , so the feature dimension of each patch is $4 \times 4 \times 3 = 48$. A

linear embedding layer is applied to this original value feature, and the vector dimension of each token is called C . Then, each input patch is passed through several self-attentive blocks (Swin Transformer Block), and the number of each token is called C . Each input patch is then passed through several self-attentive blocks (Swin Transformer Blocks), and the number of tokens per token is $H/4$, $W/4$, and together with Patch Embedding is called "Stage 1". As the network becomes more profound, the number of tokens is reduced by patch merging.

And the main job of Patch Merging is to take the neighboring small patches and merge them into one big patch so that the operation can have the effect of downsampling a feature map. The first Patch Merging layer splices each group of 2×2 adjacent patches together, and the exact positions are merged. $H/8$ and $W/8$ reduce the output resolution, and the vector dimension is changed to $2C$, which is called "Stage 2" with the Swin Transformer Block. The process is repeated twice, i.e., "stage 3" and "stage 4", with the output resolutions of $H/16$ and $W/16$, $H/32$ and $W/32$, respectively. These stages produce layered representations with the exact feature mapping resolution as typical convolutional networks (e.g., VGG [10] and ResNet [11]). Inside Swin Transformer, it contains two consecutive Blocks, each shown in Fig. 2.

The Swin Transformer is constructed by replacing the standard Multi-head Self-Attention (MSA) module in the Transformer block with a Shift Window-based Self-Attention module, leaving the other layers unchanged. As shown in Fig. 2, a Swin Transformer block consists of a shift window-based MSA module followed by a 2-layer MLP with a GELU layer. LayerNorm (LN) layers are applied before each MSA module and MLP, and residual connections are used after each module.

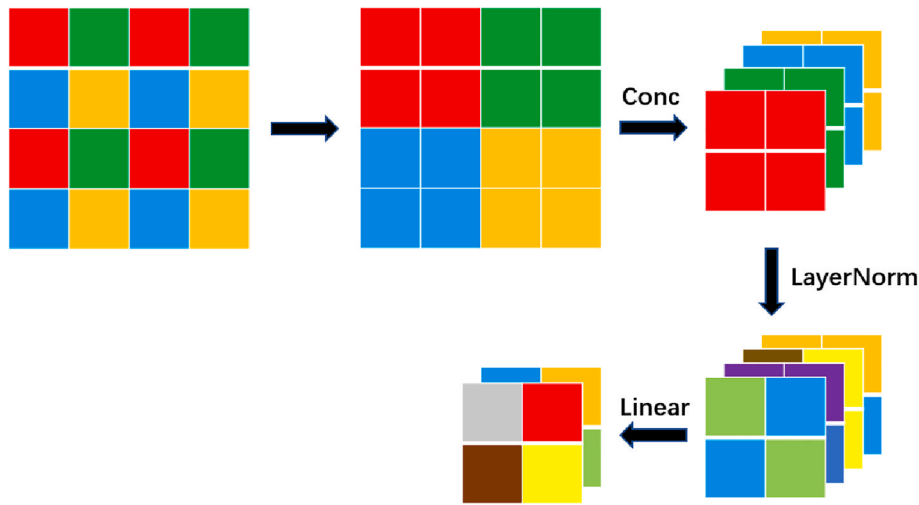


Fig. 3. Patch merging modules.

2.2. Patch Merging Modules

The Patch Merging module takes a patch of size $H \times W$, first splices it, and fully joins it in the channel dimension to form a $H/2 \times W/2 \times 4C$ feature map, then performs a Layer Normalization operation for regularization, and then performs a dimensionality reduction by linear layers to create a $H/2 \times W/2 \times 2C$ feature map. The down-sampling process of the feature map is completed. As shown in Fig. 3, the size is reduced to 1/2 of the original size, while the number of pipes is expanded to twice the original size.

Each time we do patch fusion, the four adjacent tokens are fused. Patch Merging mainly downsamples the image, similar to the pooling layer in a convolutional neural network, and primarily extracts an immense value in the region.

2.3. W-MSA modules

MSA (Multi-Head Self Attention) in the ViT network enables each pixel to be an inner product with other pixels to obtain information of all pixels by self-attentive operation, thus bringing rich global details. However, each pixel point needs to exchange information with other pixel points, which is computationally intensive and inefficient for network execution. Therefore, Swin-T divides the MSA into multiple fixed windows to form W-MSA (Windows Multi-Head Self Attention). Each pixel in a window can only be an inner product with other pixels to obtain information dramatically reduces the computation and improves the efficiency of the network.

The input is a graph, then turned into a tensor (tensor). Usually, the pipeline of the picture is three channels (RGB). After the operation of Patch Embedding, its pipeline number is 96, and the original graph's patch size is 4×4 . The pipeline is 3 (RGB), through the window partitioning, divided into several windows. After these windows are

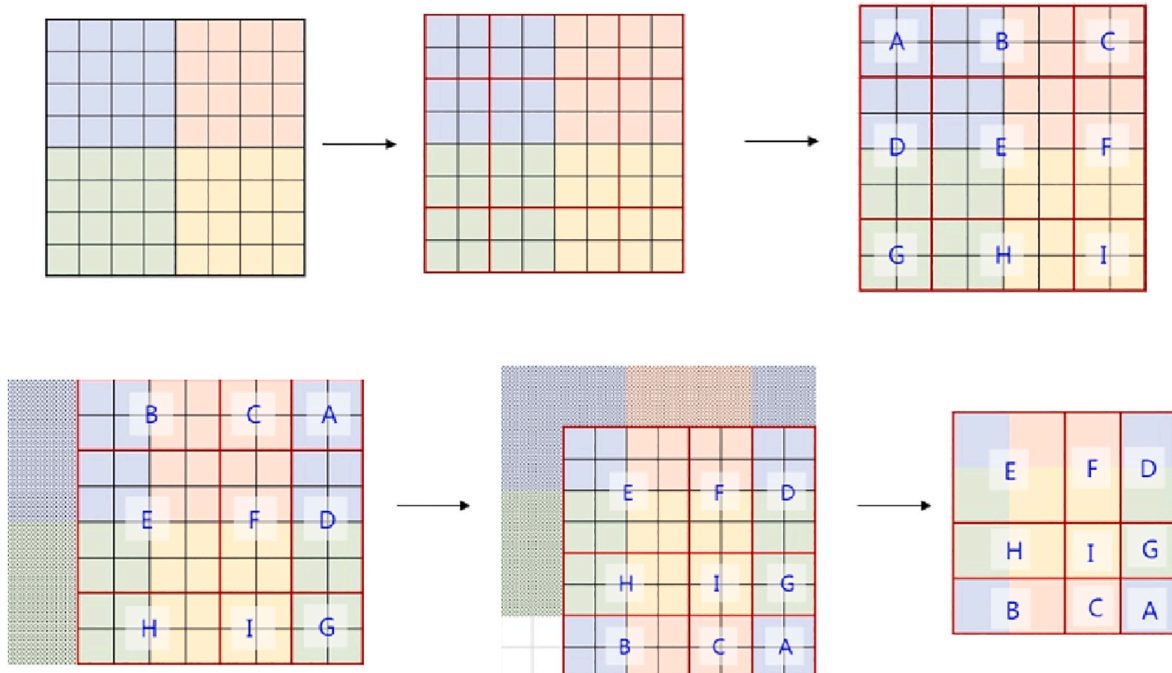


Fig. 4. Calculation method used for self-attention in shifted window partitions.

Table 1

Dataset used for the experiment.

Dataset	ID	Pictures	Training
VeRi-776	776	49,357	37,778
VehicleID	26,328	221,567	110,178

available, the self-attentiveness of each partitioned window is calculated by WMSA (Windows Multi-Head Self Attention). There is no relationship between these windows, how many tokens are input, and how many are output. The attention is calculated inside each window.

2.4. SW-MSA modules

Although W-MSA reduces the computational effort by dividing the windows, it can be seen as a reduced “field of perception” because there is no information interaction between the windows, which prevents us from obtaining more globally accurate information and thus affects the accuracy of the network. To achieve information interaction between other windows, we can slide the window, offset the window to contain different pixel points, then perform a W-MSA calculation, and connect the results of the two W-MSA calculations to combine the information contained in the pixel points of the two different windows to achieve information common between the windows. The W-MSA of the offset window constitutes the SW-MSA (Shifted Windows Multi-Head Self Attention) module, in which the window is shifted by two patches to the lower right corner of the W-MSA, forming nine blocks of different sizes. The specific structure is shown in Fig. 4. Then the nine blocks are shifted and stitched into four blocks the same size as the W-MSA using circular shifts. Masked MSA extracts the information by calculating the corresponding templates for these four spliced blocks and reversing the circular change to shift the information data patches back to their original positions. The SW-MSA mechanism completes the MSA calculation of the pixel points of the offset window and realizes the information exchange of the pixel points between different windows, thus indirectly expanding the “field of perception” of the network and improving the efficiency of information utilization.

3. Experimental verifications

This paper uses the Pytorch deep learning framework for our experiments. The experimental environment is a host computer with a 3060 graphics card with 12 GB of video memory. The parameters included in the experiments are mainly based on the settings of the Swin Transformer paper. The number of images per batch is 32, the number of windows is also divided by the original article 4*4, and the size of the input images is 256*256. Random, flip, random erase, unexpected crop, and other data enhancement strategies are used to improve the robustness of the model during training.

3.1. Comparison of datasets and methods

The two most commonly used datasets in this domain are VeRi-776 and VehicleID. The VeRi dataset [18] consists of 40,000 images of 619 vehicles captured by 20 cameras with rich annotation information, including bounding boxes, vehicle types, colors, manufacturers, etc. Each vehicle is captured by at least 2 and 18 cameras with different viewpoints, lighting, resolutions, and occlusions, constituting a highly reproducible dataset under realistic scenarios. Liu et al. [19] extended and annotated the VeRi dataset and proposed the VeRi-776 dataset, which is also one of the more widely used vehicle re-identification datasets. VehicleID is a large-scale vehicle re-identification dataset presented by Liu et al. [20], which includes 221,763 images of 26,267 vehicles, mainly containing two front and rear views. Each image has not only vehicle ID but also details of vehicle models (250 manufacturer models in total) to make the performance evaluation of the vehicle

re-identification method more comprehensive. Table 1 gives the dataset used for the experiments, containing the ID of each vehicle, the total number of images, and the number of images used for training and testing.

In this paper, we mainly compare the following methods by comparing the VAMI [22] model to solve the problem of low re-identification accuracy caused by vehicles taken at different angles and propose a multi-angle learning method. First, a single-angle image of the vehicle is input into the network to extract features, and then other view features are trained using cross-view sorting loss to obtain features for other views of the vehicle. The features generated using the network are very different from those extracted from the images retaken by the camera, so the final results are average. PROVID [12] and VSTP [13] are both multimodal retrieval models, and they encode the images of the vehicle and the temporal and spatial information associated with the images into the model, which can improve the accuracy of vehicle retrieval to some extent, and the failure to fuse the spatiotemporal features properly. The main innovation of the SSL [14] model is the use of generative adversarial networks to generate vehicle re-identification samples, thus increasing the number of vehicle re-identification training sets, and the training process of the SSL model uses a semi-supervised learning method. The RAM [15] model is a region-aware algorithm that uses four branches to form the entire vehicle re-recognition model. The four branches focus too much on local detail features and have too little ability to perceive the overall vehicle features. QD-DLF [16] extracts the features of vehicle images in horizontal, vertical, diagonal, and anti-diagonal directions, and finally, the characteristics of these methods are normalized and connected. MTCRO [17] proposes a multi-task learning method and constructs a new ranking method. TransReID is an improved method based on VIT, which mainly introduces auxiliary information embeddings to mitigate feature bias for camera/view changes by inserting learnable embeddings to merge these non-visual cues. Feature bias of camera or view changes.

3.2. Evaluation indicators

During testing, the images of the test set are retrieved and ranked using the pictures of the query set, and the closest vehicles with the same id are organized by Euclidean distance. The most widely used metrics in retrieval and re-identification tasks are Rank-N and mean average precision (mAP) to evaluate how well the model performs.

Vehicle re-identification is an image retrieval subproblem, and when measuring the performance of re-identification methods, the evaluation metrics used mainly include mAP and Rank-N table. The mean average precision is used to evaluate the overall performance of re-recognition and represents the average precision of all retrieval results. The test set is divided into query set and gallery set, and the average precision AP is first calculated for the retrieval results of each query set image q , as in Formula (2), k denotes the gallery set image serial number, n denotes the total number of galleries set images, and N denotes the total number of images of the target vehicle. $P(k)$ means the precision before the k th position in the retrieval sequence, $gt(k)$ denotes whether the k th image is the target vehicle, and Q represents the total number of query images. The final average precision of the retrieval results for all query sets is averaged, i.e., mAP , as shown in Formula (3).

$$AP = \sum_{k=1}^n P(k) * gt(k) / N \quad (2)$$

$$mAP = \sum_{q=1}^Q AP(q) / Q \quad (3)$$

The Rank-N table is used to compare the performance of different methods. If the performance difference between methods is insignificant, accurately determining good or bad performance is impossible. To reach the performance difference between other methods more

Table 2

VeRi-776 dataset comparison experiment.

Method	mAP/%	Rank-1/%
VAMI [22]	50.1	77.0
PROVID [12]	53.4	81.6
VSTP [13]	58.8	86.4
SSL [14]	61.1	88.6
RAM [15]	61.5	88.6
QD-DLF [16]	61.8	88.5
MTCRO [17]	62.6	88.0
TransReID [8]	78.2	96.5
Swin Transformer	78.6	97.3

concisely, the cumulative matching accuracy of some key matching positions is generally selected for comparison, where Rank-1 and Rank-5 are more common, indicating the probability of correctly matching the first and the fifth images in the result sequence, respectively. The evaluation index Rank-k suggests that for all the photos of the vehicle to be detected, the pictures of the matching image library are sorted in order of similarity to see whether the k th image of the sorted results is retrieved correctly, and the calculation is shown in Formula (4).

$$\text{Rank} - k = \frac{1}{m} \sum_{k=1}^m S_{i,k} \quad (4)$$

Where m denotes the total number of vehicle images to be retrieved. For the i th image, its retrieval results are sorted by similarity, and $S_{i,k} = 1$ if there are images with the same ID as the detected image in the first k images, otherwise $S_{i,k} = 0$.

3.2. Experimental results and analysis

The proposed method in this paper is compared with the algorithms with higher accuracy in recent years as follows. The results of the comparison experiments on the VeRi-776 dataset are shown in Table 2 and Fig. 5. The performance of the shipped-to Swin Transformer method is 0.6%/0.4% higher in Rank-1/mAP than the current Vision

Transformer based vehicle re-identification algorithm.

Comparative experiments on VehicleID dataset: Table 3 and Fig. 6 show the results of the VehicleID dataset compared with other methods, and Swin Transformer achieves the best performance as well. On the VehicleID dataset Rank-1/mAP outperforms the next best method TransReID by 0.4% and 0.3%, respectively.

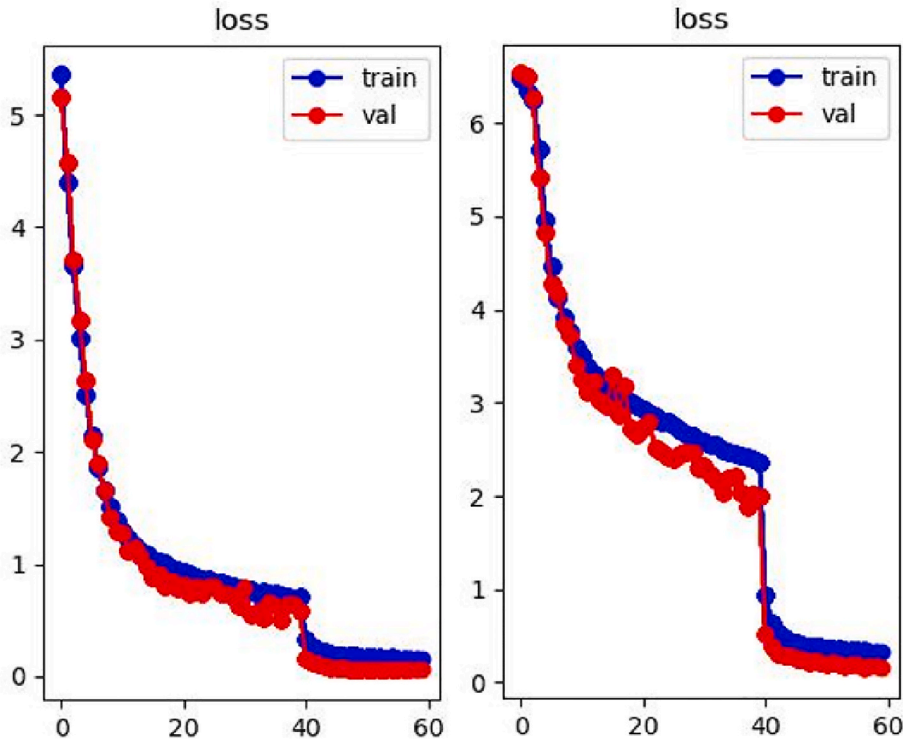
The Swin Transformer model method applied in this paper, with the above experimental results, shows that it achieves good performance on both the VeRi-776 dataset and VehicleID dataset, outperforming the current methods. This demonstrates the effectiveness and generality of the way.

The loss decreases for both datasets trained and tested during the experiments are shown in Fig. 7. The left side corresponds to the VeRi-776 dataset, and the right corresponds to the VehicleID dataset. The experiments are mainly conducted for 60 Epochs, and the loss stabilizes around the 60th time. The experimental results show that the Swin Transformer model has higher accuracy than all other models in the VeRi-776 and VehicleID datasets. The visualization results of the Swin Transformer model are shown in Fig. 8. In Fig. 8, column 1 is the image to be recognized, and columns 2 to 11 are the ten images sorted from highest to lowest in order of similarity. Columns 2 to 11 are the ten images sorted by similarity from highest to lowest. Both row 2 and row 3 show the ten results with correct matches. The last image in row 1 is an

Table 3

VehicleID dataset comparison experiment.

Method	mAP/%	Rank-1/%
DRDL [20]	49.0	73.5
C2F-Rank [21]	61.1	81.6
VAMI [22]	63.1	83.2
FDA-Net [23]	64.0	82.8
VGG + C + T + S [24]	69.9	87.3
PRN [25]	78.9	94.8
VANET [26]	83.2	95.6
TransReID [8]	83.6	97.1
Swin Transformer	83.9	97.5

**Fig. 5.** VeRi-776 dataset comparison histogram.

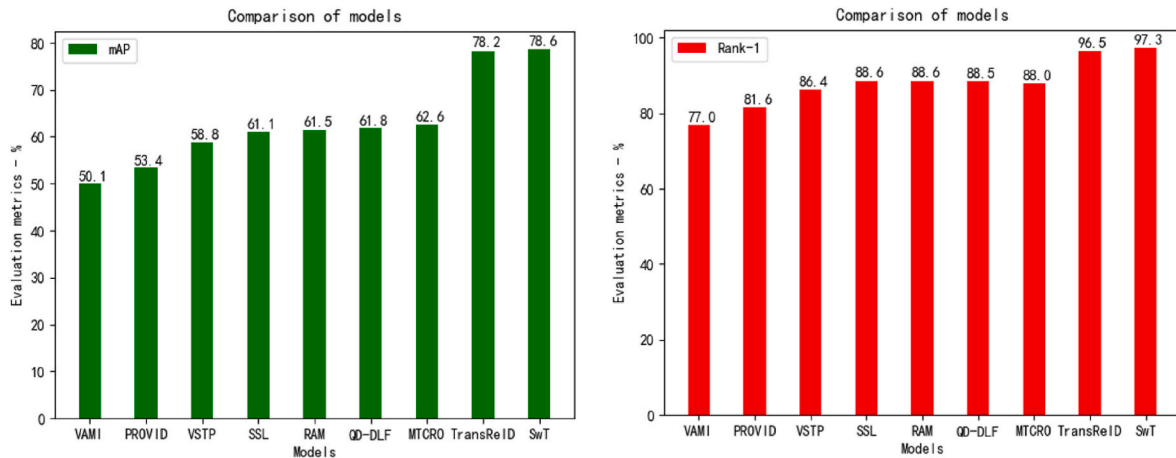


Fig. 6. VehicleID dataset comparison histogram.

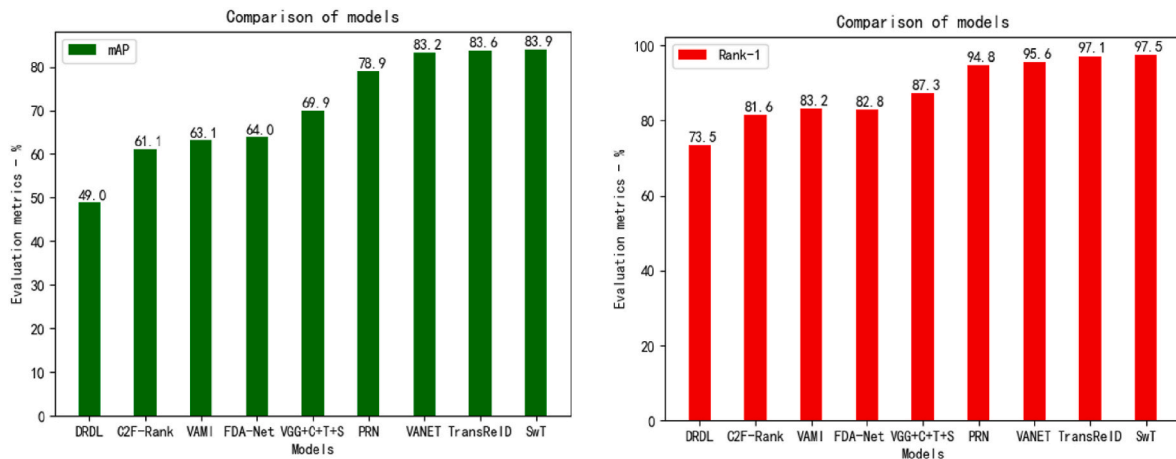


Fig. 7. Training and testing losses.



Fig. 8. Visualization of vehicle re-identification model results for Swin Transformer.

incorrect match. The visualization of the results shows that the overall appearance similarity is still relatively high, proving that Swin Transformer has a more robust feature extraction capability and accuracy in the vehicle re-identification method.

Conclusions

In this paper, we apply the Swin-Transformer model structure for vehicle re-identification and demonstrate through experiments that the recognition effect in two large vehicle datasets, VeRi-776, and

VehicleID, is significantly better than existing vehicle re-identification methods, such as VAM, FDA-Net, TransReID, etc. As the backbone network for feature extraction, the method outperforms the comparison methods in both Rank-1 and mAP, effectively improves the vehicle retrieval accuracy of the vehicle re-identification model, and can make the association between features closer, enhance the learning of robust features, and improve the feature robustness and discriminatory performance. It provides a specific reference for the research of vehicle re-identification algorithms.

Credit author statement

Jianrong Li: Conceptualization, Software, **Chang Yu:** Methodology, Software, Writing - Original Draft, **Jinyuan Shi:** Writing - Review & Editing, Visualization, **Chuanlei Zhang:** Writing - Review & Editing, **Ting Ke:** Writing - Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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